SHIFT: Scribble-Driven Hybrid Iterative Feature Tracker for Multimodal Remote Sensing Image Change Detection

Yongjie Zheng, *Student Member, IEEE*, Sicong Liu, *Senior Member, IEEE*, and Lorenzo Bruzzone, *Fellow, IEEE*

***Abstract*—Multimodal remote sensing change detection (MRS-CD) is essential for many geospatial applications. However, it remains challenging due to spectral heterogeneity, temporal misalignment, and inconsistencies between different data sources. Existing MRS-CD methods, particularly those employing graph structures, often suffer from parameter sensitivity, limited generalization across datasets, and difficulty in detecting fine-grained changes. Meanwhile, current weakly supervised approaches typically rely on image-level annotations, which often result in coarse localization, blurred boundaries, and limited transferability across heterogeneous modalities of changes. In this work, we present scribble-driven hybrid iterative feature tracker (SHIFT), a weakly supervised MRS-CD approach guided by sparse scribble annotations. SHIFT employs a scribble-driven feature extraction module to extract multimodal features, converting scribble annotations into smooth attention maps with a learnable Gaussian blur module to focus the model on potential change areas. Through iterative refinement, SHIFT enhances detection confidence in change regions. Experiments on 13 public datasets demonstrate the superior performance of the proposed approach compared with the number of state-of-the-art (SOTA) methods. Our approach achieves high accuracy with minimal annotations and robust generalization across diverse multimodal scenarios. The source code will be made publicly available at https://github.com/MissYongjie/SHIFT**

***Index Terms*—Change detection, heterogeneous, homogeneous, multimodal image analysis, remote sensing, scribble, weakly supervised.**

I. Introduction

IN GEOSPATIAL analysis, multimodal remote sensing change detection (MRS-CD) plays a crucial role in applications, such as environmental monitoring, disaster damage assessment, and urban development [1]. By integrating data captured from heterogeneous sensors, such as optical, infrared, and SAR, MRS-CD enables the identification of temporal changes with enhanced semantic and structural awareness. This capability is particularly important in scenarios, where acquiring temporally aligned, high-quality imagery from a single sensor is constrained by satellite revisit cycles, weather conditions, or acquisition intervals [2], [3], [4].

Despite its advantages, MRS-CD remains inherently challenging. Specifically, spectral heterogeneity, spatial resolution mismatches, and temporal misalignment across modalities significantly hinder pixelwise correspondence and increase the complexity of cross-domain modeling [5]. These challenges are further amplified in supervised learning settings, where the acquisition of dense annotations remains costly and labor-intensive due to the requirement for expert expertise [6], [7], [8].

To alleviate annotation burdens, recent studies have increasingly turned to unsupervised, semi-supervised, or weakly supervised frameworks. These paradigms aim to relax the reliance on dense labels while maintaining acceptable performance levels. For example, recent semi-supervised approaches, such as Confucius Tri-Learning [9], have exploited both positive and negative examples to improve robustness under limited annotations. Meanwhile, in unsupervised settings, structure-based methods adopting superpixel segmentation techniques (such as SLIC [10]) have shown promise in preserving semantic consistency across modalities. These methods segment input images into homogeneous regions and extract structure-aware features, which are then processed by regression models or deep learning backbones such as U-Net or GANs [5], [11], [12]. However, their effectiveness is frequently constrained by hyperparameter sensitivity (e.g., superpixel granularity), which can limit generalization across diverse scenes and hinder the detection of fine-grained or small-scale changes [13]. This effect is visually illustrated in Fig. 1, where different superpixel scales result in varying region representations across time-series images.

Another relevant research direction focuses on high-resolution remote sensing (HR-RS) tasks, particularly building-level change detection. Benchmarks, such as LEVIR-CD and WHU-CD, have popularized the use of image-level annotations (e.g., change/no change) over very-HR-RS images. These labels are then projected to the pixel level via class activation maps (CAMs), enabling weakly supervised training.

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Yongjie Zheng and Lorenzo Bruzzone are with the Department of Information Engineering and Computer Science, University of Trento, 38123 Trento, Italy (e-mail: yongjie.zheng@unitn.it; lorenzo.bruzzone@unitn.it).

Sicong Liu is with the College of Surveying and Geo-Informatics, Tongji University, Shanghai 200092, China (e-mail: sicong.liu@tongji.edu.cn).

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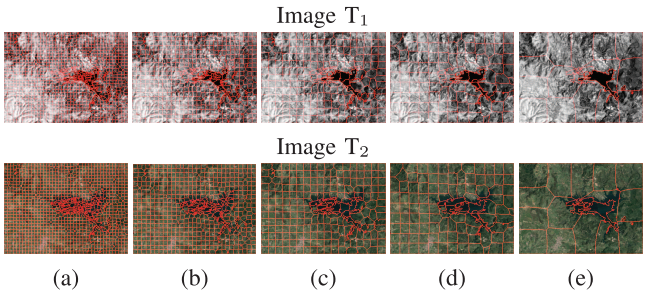


Fig. 1. Comparison of superpixel co-segmentation maps across varying scales for Case #1 (it corresponds to the flooding event on HetData #1). Superpixels are generated using the Gaussian mixture model-based superpixel (GMMSP) segmentation method [11]. The scales represent (a) 1/100, (b) 1/200, (c) 1/500, (d) 1/1000, and (e) 1/5000 of the total pixel count.

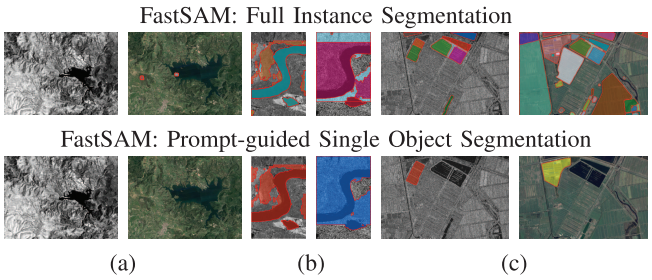


Fig. 2. Segmentation results of FastSAM [16] [(Top row) Without prompts and (bottom row) with prompts)] on three change detection cases. (a) Case #1, (b) Case #2, and (c) Case #3, each with its corresponding T_1 and T_2 images.

Despite reducing annotation cost, this strategy typically suffers from imprecise localization, as CAMs tend to highlight only the most salient regions and overlook object boundaries or subtle changes.

In this context, advances in computer vision (CV) have introduced vision foundation models such as the segment anything model (SAM) [14], [15] and FastSAM [16], which facilitate prompt-driven segmentation from sparse annotations (e.g., points, boxes, or scribbles). While these models demonstrate remarkable performance on natural images at submeter resolutions, their application to medium- and low-resolution remote sensing imagery (e.g., 5–30 m/pixel) has been less successful. As illustrated in Fig. 2, FastSAM yields suboptimal results on multimodal datasets, revealing a notable performance gap when transferring these models to remote sensing contexts. This discrepancy arises from fundamental differences in spatial texture, spectral complexity, and scene semantics, underscoring the need for domain-specific solutions.

Despite progress across various paradigms, several critical limitations remain unresolved in the field of MRS-CD.

1) *Unsupervised MRS-CD Methods:* While free from annotation requirements and thus highly scalable, these methods tend to rely heavily on predefined region aggregation strategies, such as superpixel over-segmentation or graph-based modeling. These strategies often blur spatial boundaries and overlook fine-grained changes.
2) *Weakly Supervised Approaches:* Most of these methods based on image-level labels or CAM-derived attention maps are typically applied to high-resolution settings. When transferred to heterogeneous or lower resolution modalities, they often suffer from unclear semantics and imprecise boundary localization, which limits their generalization across diverse modalities.
3) *Scribble-Based Models:* Although promising in interactive CV-domain segmentation tasks, these methods have not been effectively adapted to RS-CD. Most existing models treat scribbles as hard labels or predefined masks, lacking mechanisms to fully exploit their flexibility for region-level guidance or iterative refinement.

These limitations highlight a key challenge in MRS-CD: the absence of a supervision strategy capable of simultaneously ensuring annotation efficiency, spatial precision, and generalizability across heterogeneous modalities. While existing methods employ various supervision levels, they frequently exhibit either inadequate localization or limited scalability. Scribble annotations present a promising compromise by delivering dense spatial cues at low annotation cost. However, their potential in MRS-CD remains underexplored, particularly regarding their utilization as dynamic supervision signals rather than static labels.

To address the aforementioned limitation, we propose a scribble-driven hybrid iterative feature tracker (SHIFT), which is a weakly supervised MRS-CD framework that integrates sparse scribble annotations during both training and postprocessing stages. SHIFT incorporates a learnable attention mechanism to transform sparse inputs into scalable and accurate CD guidance. Our novel contributions include the following.

1) *Scribble-Guided Multimodal Change Detection:* We propose the first MRS-CD framework that fully integrates scribble annotations into both training and postprocessing stages, enabling accurate detection with minimal supervision.
2) *Learnable Gaussian Attention Mechanism:* We introduce a trainable Gaussian blur module that adaptively transforms scribble inputs into region-level attention maps, improving focus and enhancing fine-grained change localization.
3) *Consistency and Separation Constraints:* We design novel loss functions that promote intraclass feature consistency in unchanged areas and interclass separation in changed regions, leading to improved localization accuracy and robustness.

The remainder of this article is organized as follows. Section II reviews related work on MRS-CD. Section III details our proposed SHIFT. Section IV presents experimental results and the related analyses. Finally, Section V concludes this article and discusses future work.

II. Related Works

Traditional fully supervised change detection methods achieve state-of-the-art (SOTA) performance but heavily rely on expensive pixel-level annotations, making them impractical for large-scale and multimodal remote sensing datasets. Consequently, alternative paradigms, including semi-supervised, weakly supervised, and unsupervised methods, have emerged

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TABLE I
COMPARISON OF LEARNING PARADIGMS IN REMOTE SENSING CHANGE DETECTION

Key Characteristics	Unsupervised Learning	Semi-Supervised Learning	Weakly Supervised Learning
Label Requirement	No labels	Few pixel-level labels + many unlabeled	Coarse labels (e.g., image-level, scribbles)
Sample Format	Paired images	Paired images + partial masks	Paired images + sparse annotations
Typical Methods	Structural regression, graph learning	GANs, consistency regularization	CAM refinement, pseudo-labeling
Advantages	Label-free, scalable	High accuracy with reduced cost	Cost-effective for sparse supervision
Limitations	Limited accuracy	Label-ratio sensitive	Coarse change localization

to alleviate annotation constraints, facilitating broader applicability and scalability. Unsupervised CD (USCD) methods, especially in multimodal CD, leverage inherent structural consistencies without explicit labeling. Semi-supervised CD (SSCD) approaches utilize a small set of labeled samples alongside a large set of unlabeled data to improve model generalization. Meanwhile, weakly supervised CD (WSCD) approaches utilize affordable image-level, point-level, and patch-level annotations are becoming increasingly important, particularly in high-resolution tasks such as building change detection. To better clarify the distinctions among these learning paradigms, Table I summarizes their key characteristics, including advantages and limitations, in the context of RS-CD.

A. Unsupervised Remote Sensing Change Detection

Due to differences in spectral, spatial, and temporal resolutions, achieving effective comparison between multimodal images remains challenging. Early unsupervised methods primarily employed image regression and domain adaptation techniques to mitigate inherent disparities across modalities. Notable examples include affinity matrix difference guided image regression (AMD-IR) [17] and conditional GAN (CGAN)-based transformations [18], which align multimodal data in a unified domain. However, these techniques primarily focus on radiometric similarities and often fail to handle structural inconsistencies effectively.

Graph-based methods have recently gained prominence due to their ability to preserve structural details. Techniques, such as NPSG [19], GIR-MRF [20], and SDIR [21], employ graphs for spatial and structural alignment, demonstrating robustness to noise. Yet, structural asymmetry remains a significant limitation for these approaches. Advanced frameworks like structural regression fusion (SRF) [22] further improve alignment by integrating forward and backward transformations, thereby effectively addressing both similarities and differences.

Further improvements in unsupervised approaches involve leveraging adaptive strategies for multimodal feature representation. For instance, graph convolutional autoencoders (e.g., SR-GCAE [23], SDC-GAE [2]) iteratively refine multimodal features through structural and spectral self-similarities. CNN-based XNet and ACE-Net [24] focus on structure-preserving transformations to enhance robustness across diverse scenarios. Despite advancements, these methods inherently suffer from limited accuracy due to their dependence on intrinsic image features, particularly structural differences, thereby restricting precise change detection.

B. Semi-Supervised Remote Sensing Change Detection

SSCD methods pay more attention to the efficient usage of annotation-free data in order to achieve satisfactory results with limited labeled data alongside abundant unlabeled data. Recent SSCD approaches generally fall into two categories: adversarial learning-based methods and consistency learning-based methods [25], [26]. Adversarial methods, such as SemiCDNet [27], utilize GANs to generate change predictions for unlabeled data, aiming to minimize differences from labeled predictions. However, these methods often suffer from instability due to the complexity of adversarial optimization strategies. Consistency learning-based methods, like revisiting consistency regularization (RCR) [28], focus on ensuring prediction invariance across perturbations, including feature and prediction map disturbances. Nevertheless, these perturbations often lack explicit physical interpretations, limiting the insights into model optimization.

To address these limitations, Zhang et al. [29] proposed the feature-prediction alignment (FPA) framework. FPA introduces two alignment strategies: class-aware feature alignment (FA), which aligns features across regions according to change/no-change classes, and pixelwise prediction alignment (PA), aligning predictions from strongly augmented images with pseudo-labels derived from weak augmentations. Despite recent progress, SSCD methods still face key challenges. Their performance is often sensitive to the proportion and quality of labeled data and struggles under heterogeneous or low-resolution settings. Moreover, current approaches lack effective mechanisms to handle semantic inconsistencies and noisy supervision, limiting their robustness in real-world applications.

C. Weakly Supervised Remote Sensing Change Detection

To alleviate the dependence on labor-intensive pixel-level annotations, WSCD has emerged as a promising alternative by leveraging coarse supervision forms, predominantly image-level labels (e.g., change/no change), and to a lesser extent, patch-level or region-level cues. These methods typically rely on CAMs, pseudo-label refinement, and consistency constraints to infer accurate pixel-level predictions.

A representative stream of work is based on CAM localization and its enhancement. For instance, ESAM-CD [30] integrates multiscale CAM fusion with EfficientSAM, fine-tuned via a low-rank adaptation (LoRA) strategy, to improve pseudo-label quality and boundary delineation. CS-WSCDNet [31] combines CAM localization with the prompt-free segmentation capabilities of the SAM for pixelwise supervision.

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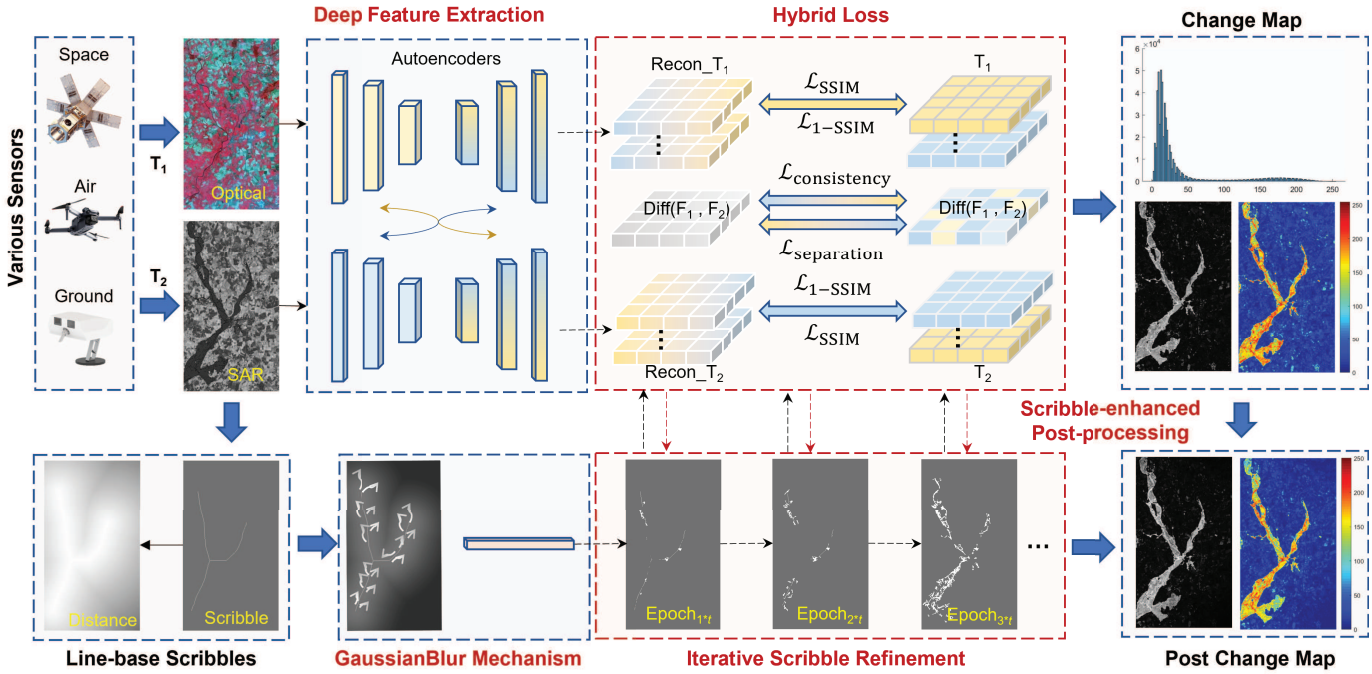


Fig. 3. Overview of SHIFT. Starting from multimodal deep feature extraction, our approach employs a Gaussian blur attention mechanism, a hybrid loss with iterative scribble refinement, and a postprocessing with morphological filtering and scribble-guided refinement to generate the final change map.

Transformer-based methods, such as ACWCD [32], exploit the inherent attention mechanisms of ViT backbones to refine CAMs via attention-guided modulation. SSCMNet [33] further introduces a multiscale self-supervised constraint and change-mask guidance module to reduce noise and preserve object completeness in weak annotations. Other methods, such as TransWCD [34] and DISep [35], propose decoder regularization or instance separation strategies to bridge the gap between coarse labels and dense outputs.

Complementing discriminative methods, hybrid generative-discriminative frameworks further improve supervision robustness. FCD-GAN [36] unifies multiple supervision paradigms within a fully convolutional adversarial framework. KD-MSI [37] introduces a teacher–student distillation mechanism guided by CAM-derived knowledge. DeepCut [38] refines interactive GrabCut masks using deep semantic features, while TICDN [38] enhances temporal invariance through adversarial learning. Guided anisotropic diffusion [39] serves as an edge-aware regularizer to boost spatial consistency during iterative training.

In addition, MS-Former [40] demonstrates the effectiveness of memory-supported transformers under patch-level supervision, leveraging bidirectional attention for fine-grained change localization. Despite these advancements, current WSCD methods remain largely constrained to high-resolution building scenarios and often struggle with semantic gaps in multimodal settings.

III. Proposed Method

To address the challenges of obtaining representative training samples for CD due to large modality gaps in MRS images and the reliance of existing MRS-CD methods on graph-based unsupervised translation, we introduce a simple yet effective scribble-based annotation approach. This approach avoids over dependence on training samples while still providing targeted regions of interest. As illustrated in Fig. 3, our approach consists of five main components: 1) scribble annotations; 2) multimodal deep feature extraction; 3) Gaussian blur attention mechanism; 4) hybrid loss and iterative scribble refinement; and 5) postprocessing mechanism with morphological filtering and scribble-guided refinement. The complete SHIFT process is summarized in Algorithm 1. Further details are provided as follows.

A. Scribble Annotations

In this work, the sparse scribble annotations are adopted to provide weak supervision for multimodal change detection tasks. Specifically, annotators are instructed to draw free-hand lines over the approximate locations where changes are observed, directly on the "before" and "after" image pairs. These scribbles are class-specific and only indicate "positive change regions," without any labeling of unchanged areas.

The annotations can be created using a simple drawing tool that supports freehand input. Annotators are not required to outline object boundaries; instead, they loosely trace over representative portions of the changed regions. This greatly reduces the annotation burden while still providing informative cues for training.

The shape and length of each scribble are not strictly constrained, allowing flexibility in annotation. However, to ensure effective supervision, important guidelines are to draw each scribble within the changed region and to make it smaller than the full extent of the object. This strategy ensures that even if the annotations are coarse, they remain fully enclosed by the actual change and can reliably guide the model. In

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Algorithm 1 SHIFT

Input: Input images $\mathbf{X}_1$, $\mathbf{X}_2$; Initial scribble $\mathbf{S}_0$; Target coverage τ
Output: Final binary change mask $\mathbf{B}_{final}$
Initialize: Model $\mathcal{M}$ with dual encoders, decoders, Gaussian blur attention module;
Initialize optimizer $\mathcal{O}$ and scheduler $\mathcal{S}$;
Set scribble $\mathbf{S} \leftarrow \mathbf{S}_0$;
for $e = 1$ **to** E **do**
  Forward pass: $[\mathbf{F}_1, \mathbf{F}_2, \hat{\mathbf{X}}_1, \hat{\mathbf{X}}_2, \mathbf{A}] = \mathcal{M}(\mathbf{X}_1, \mathbf{X}_2, \mathbf{S})$
  Compute feature distance map: $\mathbf{D} = \|\mathbf{F}_1 - \mathbf{F}_2\|_2$
  Compute total loss: $\mathcal{L} = \alpha\mathcal{L}_{\text{con}} + \beta\mathcal{L}_{\text{sep}} + \gamma\mathcal{L}_{\text{ssim}}$;
  Update model parameters: optimizer $\mathcal{O}$ and scheduler $\mathcal{S}$;
  if e mod $T == 0$ **then**
    Compute coverage: Coverage $= \frac{\sum \mathbf{S}}{H \times W}$;
    if *Coverage* $< \tau$ **then**
      Dilate $\mathbf{S}$ to define candidate region;
      Select top-k% high-distance pixels from $\mathbf{D}$;
      Generate updated scribble $\mathbf{S}_{\text{new}}$;
      Assign $\mathbf{S} \leftarrow \mathbf{S}_{\text{new}}$;
Post-processing:
Compute inverse distance map $\mathbf{W}_{\text{inv}}$ from final scribble $\mathbf{S}$;
Compute weighted difference map $\mathbf{D}_w = \mathbf{D} \cdot \mathbf{W}_{\text{inv}}$;
Apply adaptive thresholding to obtain binary map $\mathbf{B}$;
Apply morphological filtering to refine $\mathbf{B}$ into $\mathbf{B}_{final}$;
return $\mathbf{B}_{final}$

general, more precise scribbles offer better guidance, but the method remains robust under imprecise inputs as long as the scribbles roughly localize the change.

B. Multimodal Deep Feature Extraction

To address the inherent challenges of aligning and extracting robust features from MRS images with different spectral bands and resolutions, we introduce a multimodal deep feature extraction module. This module takes as input two co-registered multimodal images, $\mathbf{X}_1 \in \mathbb{R}^{H \times W \times C_1}$ and $\mathbf{X}_2 \in \mathbb{R}^{H \times W \times C_2}$, captured over the same geographic area at times t_1 and t_2, respectively. Here, H and W denote spatial dimensions, while C_1 and C_2 represent the channel or spectral band counts.

Both $\mathbf{X}_1$ and $\mathbf{X}_2$ images are processed through a dual-branch encoder–decoder architecture. The encoder maps each input to a shared feature space, producing deep feature representations $\mathbf{F}_1 = \text{Encoder}(\mathbf{X}_1)$ and $\mathbf{F}_2 = \text{Encoder}(\mathbf{X}_2)$, where cross-modal comparisons can be effectively conducted. These encoded features are then reconstructed via decoders, yielding $\hat{\mathbf{X}}_1 = \text{Decoder}(\mathbf{F}_1)$ and $\hat{\mathbf{X}}_2 = \text{Decoder}(\mathbf{F}_2)$. The deep feature representations $\mathbf{F}_1$ and $\mathbf{F}_2$ as well as $\mathbf{X}_1$ and $\mathbf{X}_2$ will contribute to hybrid loss calculations and drive the training of the network. As explained in Section III-D, the change map is generated by normalizing the feature distance between $\mathbf{F}_1$ and $\mathbf{F}_2$, providing an initial difference map for subsequent analysis.

C. Gaussian Blur Attention Mechanism

As mentioned before, MRS-CD images captured at different times often exhibit discrepancies in style, resolution, and temporal characteristics, creating an information gap where nonrelevant differences may be confused with genuine changes. To address this, we introduce an initial sparse annotation using simple scribbles to roughly localize potential change areas. Here, a sparse binary scribble annotation map $\mathbf{S} \in \mathbb{R}^{H \times W}$ is provided, with $\mathbf{S}(i, j) = 1$ marking potential change regions. This guides the feature extraction process, focusing on regions of interest.

To enhance these sparse annotations iteratively, we propose a Gaussian Blur attention mechanism. This mechanism uses a learnable Gaussian kernel to transform the binary scribble map $\mathbf{S} \in \mathbb{R}^{H \times W}$ into a smooth, continuous attention map $\mathbf{A} \in \mathbb{R}^{H \times W}$, focusing on probable change regions

$$\mathbf{A} = \mathbf{S} * \mathbf{G}_\sigma \tag{1}$$

where $*$ represents the convolution and σ is a learnable parameter. The learnable σ allows the model to control the spread of attention dynamically, capturing context around annotated regions.

This attention map $\mathbf{A}$ is then applied elementwise to the extracted feature representations $\mathbf{F}_1$ and $\mathbf{F}_2$, producing focused features

$$\mathbf{F}_1' = \mathbf{F}_1 \odot \mathbf{A}, \quad \mathbf{F}_2' = \mathbf{F}_2 \odot \mathbf{A} \tag{2}$$

where $\odot$ denotes the elementwise multiplication. This operation emphasizes relevant change areas while down-weighting irrelevant ones.

The Gaussian blur attention mechanism provides three benefits: 1) dynamic focus—by adjusting attention based on σ, the model ones in on probable change regions; 2) smooth localization—it creates a gradual transition around annotations, enhancing spatial coherence; and 3) iterative refinement—as detailed in Section III-D, this mechanism is iteratively refined to progressively concentrate on high-confidence change regions.

Unlike conventional Gaussian blur methods that rely on fixed kernels, our module incorporates a learnable standard deviation σ, enabling adaptive control over the spatial spread of attention. This flexibility allows the model to better align with the underlying characteristics of the scene and the provided scribble annotations. During training, the attention maps are progressively refined, leading to improved localization accuracy and effective suppression of irrelevant background noise. To the best of our knowledge, this work presents the first integration of a trainable Gaussian blur mechanism within a weakly supervised MRS-CD framework, enhancing both adaptability and detection performance.

D. Hybrid Loss and Iterative Scribble Refinement

To progressively refine the change map while preserving spatial coherence, we propose a hybrid loss combined with an iterative automatic scribble update mechanism. These components work together to optimize the model's ability to identify high-confidence change regions with minimal supervision, gradually transforming sparse scribble annotations into dense and reliable change maps. This process is visualized in Fig. 4.

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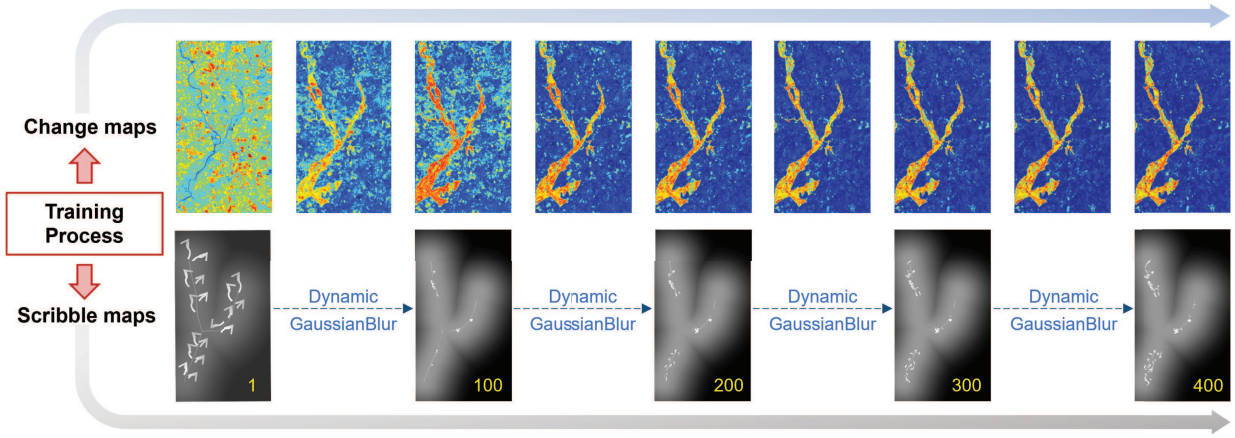


Fig. 4. Training process of the proposed SHIFT. (Row 1) Change maps generated by the feature extraction module during training. (Row 2) Iterative update of scribble maps using a dynamic Gaussian blur mechanism.

1) Hybrid Loss: Hybrid loss integrates three essential components: feature consistency loss, feature separation loss, and SSIM loss.

Feature Consistency Loss. This term penalizes differences in unchanged regions. Given the deep feature representations $\mathbf{F}_1$ and $\mathbf{F}_2$ of the input images, the consistency loss promotes similarity in areas marked as unchanged by the attention map $(1 - \mathbf{A})$. It is defined as follows:

$$\mathcal{L}\text{con} = \frac{\sum_{i,j} (1 - \mathbf{A}(i,j)) |\mathbf{F}_1(i,j) - \mathbf{F}_2(i,j)| 2}{\sum_{i,j} (1 - \mathbf{A}(i,j)) + \epsilon} \tag{3}$$

where (i, j) refers to the position of a pixel and ϵ prevents the division by 0.

Feature Separation Loss. To enhance contrast in change regions, this loss enforces a margin between features in these areas, weighted by $\mathbf{A}$. It is defined as follows:

$$\mathcal{L}\text{sep} = \frac{\sum_{i,j} \mathbf{A}_{ij} \cdot \max\left(0, m - \left|\mathbf{F}_1^{ij} - \mathbf{F}_2^{ij}\right|\right)^2}{\sum_{i,j} \mathbf{A}(i,j) + \epsilon}. \tag{4}$$

This encourages distinct differences in changed regions, improving the model's ability to recognize true changes.

SSIM Loss. SSIM loss promotes structural consistency by reducing differences between reconstructed and original images. It includes four SSIM comparisons: a) $\hat{\mathbf{X}}_1$ with $\mathbf{X}_1$; b) $\hat{\mathbf{X}}_2$ with $\mathbf{X}_2$; c) $\hat{\mathbf{X}}_1$ with $\mathbf{X}_2$; and d) $\hat{\mathbf{X}}_2$ with $\mathbf{X}_1$. Maximizing SSIM within matched pairs and minimizing it across mismatched pairs enhances spatial coherence.

The total loss is defined as follows:

$$\mathcal{L}\text{total} = \alpha \cdot \mathcal{L}\text{con} + \beta \cdot \mathcal{L}\text{sep} + \gamma \cdot \mathcal{L}_{\text{SSIM}} \tag{5}$$

where α, β, and γ are the hyperparameters that balance each term's contribution.

2) Scribble Update Mechanism: To refine the capability of the model to focus on high-certainty change regions, we employ an iterative scribble update mechanism. First, we calculate the coverage ratio of the current scribble annotations as Coverage $= \left(\sum \mathbf{S}/H \times W\right)$. If coverage reaches a predefined target, the updating process stops. Otherwise, the model generates a feature distance map $d(i, j) = \|\mathbf{F}_1(i, j) - \mathbf{F}_2(i, j)\|_2$, highlighting probable change regions where high values indicate large feature differences.

Using this distance map, we automatically expand the scribble annotations by adding high-confidence pixels in the vicinity of existing scribbles. This is achieved by dilating $\mathbf{S}$ and selecting pixels from the distance map above a threshold within the dilated area

$$\mathbf{S}_{\text{new}} = \text{Dilate}(\mathbf{S}) \wedge (d(i,j) > \text{Threshold}). \tag{6}$$

The updated scribble $\mathbf{S}_{\text{new}}$ is reintroduced into the training loop, allowing the model to iteratively refine its attention to change regions with increased accuracy.

E. Postprocessing Mechanism

To enhance localization accuracy and refine the initial change map, we employ a postprocessing mechanism comprising three steps. In the first step, we generate a weighted difference map, $\mathbf{W}_{\text{diff}}$, using the inverse distance transform of the scribble annotations, which assigns higher weights to pixels near the scribbles to strengthen confidence in these regions and suppress noise in distant areas. In the second step, an adaptive thresholding technique is applied to $\mathbf{W}_{\text{diff}}$, creating a binary change map $\mathbf{B}_{\text{change}}$ that identifies potential change regions with flexibility. In the final step, we apply region-specific morphological filtering, defining close, medium, and far zones based on scribble proximity and employing customized kernel sizes for each zone [42]. This ensures smooth boundaries, reduces noise, and retains prominent connected regions, resulting in a refined spatially coherent change map.

IV. Experiments

A. Description of Datasets

To better validate the generalization and robustness of the proposed SHIFT approach, we comprehensively collected 13 datasets, categorized into two distinct groups: Heterogeneous Datasets (HetData) [see Fig. 5(a)–(d)] and Homogeneous Datasets (HomoData) [see Fig. 6(a)–(d)]. These datasets include diverse sensor types (optical and SAR), spatial resolutions, geographical regions, and change scenarios. Details are as follows:

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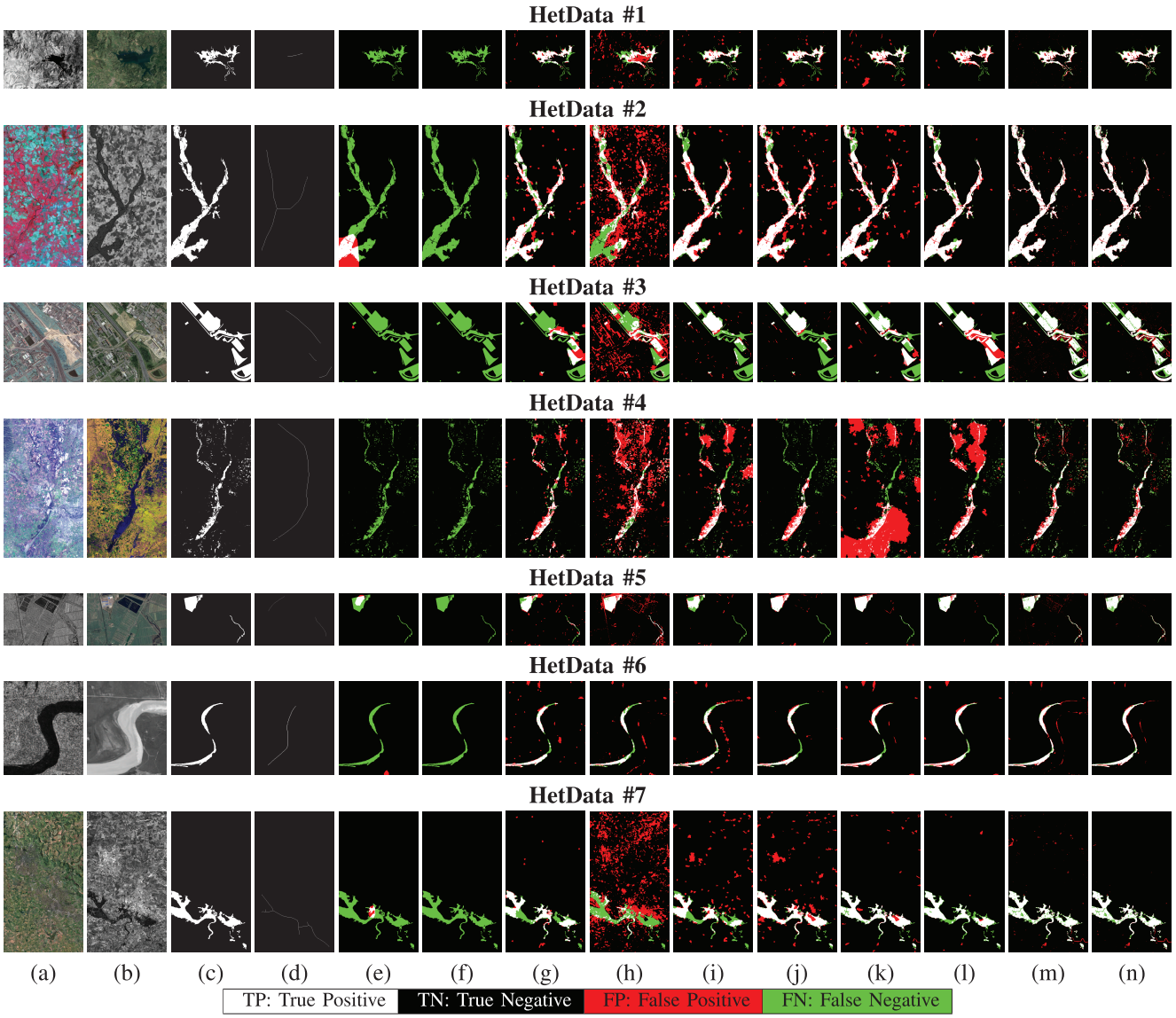


Fig. 5. Comparison of experimental results obtained by different methods on the considered heterogeneous datasets (HetData #1–7). Each row represents a dataset. (a) T_1 Image. (b) T_2 Image. (c) Ground-truth (GT) map. (d) Scribble map. Change detection maps obtained by (e) TransWCD, (f) ACWCD, (g) IRG-McS, (h) NPSG, (i) SCASC, (j) GIR-MRF, (k) SDIR, (l) SRF, (m) proposed SHIFT (Base), and (n) proposed SHIFT (Post).

TABLE II

DESCRIPTION OF HETEROGENEOUS (HET) AND HOMOGENEOUS (HOMO) DATASETS [41]

Dataset		Sensor	Size (pixels)	Date	Location	Event (& Spatial resolution)
Het	#1	Landsat-5 / Google Earth	300 × 412 × 1 (3)	Sept. 1995 - July 1996	Sardinia, Italy	Lake expansion (30m)
	#2	Spot / NDVI	990 × 554 × 3 (1)	1999 - 2000	Gloucester, England	Flooding (≈ 25m)
	#3	Pleiades / WorldView2	2000 × 2000 × 3 (3)	May 2012 - July 2013	Toulouse, France	Construction (0.52m)
	#4	Landsat-8 / Sentinel-1A	875 × 500 × 3 (3)	Jan. 2017 - Feb. 2017	California, USA	Flooding (≈ 15m)
	#5	Radarsat-2 / Google Earth	593 × 921 × 1 (3)	June 2008 - Sept. 2012	Shuguang Village, China	Building construction (8m)
	#6	Radarsat-2 / Google Earth	343 × 291 × 1 (3)	June 2008 - Sept. 2010	Yellow River, China	Embankment change (8m)
	#7	QuickBird 2 / TerraSAR-X	4135 × 2325 × 3 (1)	July 2006 - July 2007	Gloucester, England	Flooding (0.65m)
Homo	#1	Google Earth / Google Earth	500 × 500 × 3 (3)	Sept. 2012 - Mar. 2013	Beijing, China	Construction (1m)
	#2		500 × 500 × 3 (3)			
	#3	Radarsat-2 / Radarsat-2	289 × 257 × 1 (1)	June 2008 - June 2009	Yellow River Estuary, China	Farmland irrigation (8m)
	#4		280 × 450 × 1 (1)			
	#5		444 × 291 × 1 (1)			
	#6		291 × 306 × 1 (1)			

1) Heterogeneous Datasets: Heterogeneous datasets primarily focus on regions affected by natural disasters, where acquiring homogeneous data during critical periods is often challenging. These datasets include two distinct multimodal

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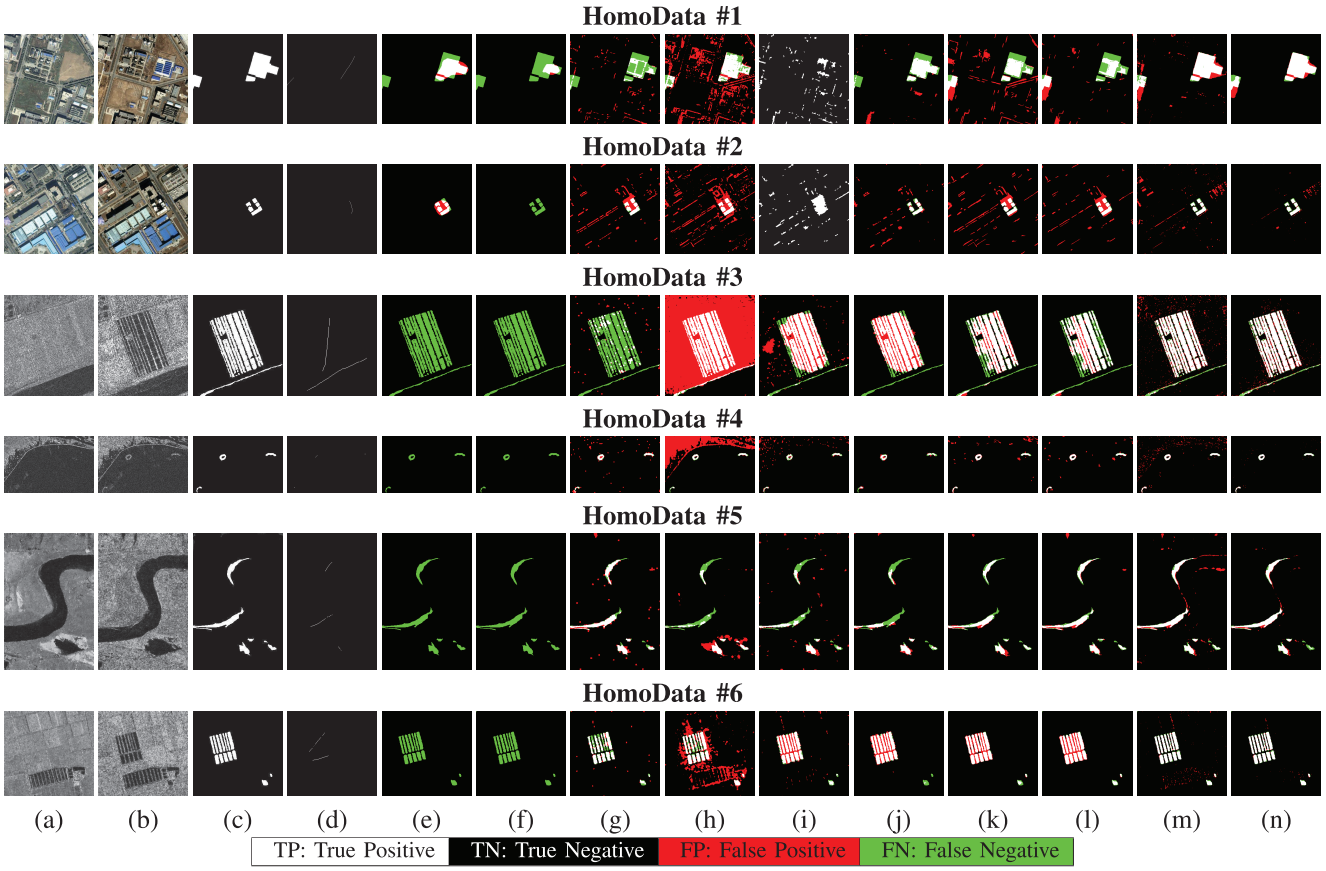


Fig. 6. Comparison of experimental results obtained by different methods on the considered homogeneous datasets (HomoData #1–6). Each row represents a dataset. (a) T_1 Image. (b) T_2 Image. (c) Ground-truth (GT) map. (d) Scribble map. Change detection maps obtained by (e) TransWCD, (f) ACWCD, (g) IRG-McS, (h) NPSG, (i) SCASC, (j) GIR-MRF, (k) SDIR, (l) SRF, (m) proposed SHIFT (Base), and (n) proposed SHIFT (Post).

scenarios: a) the cross-sensor scenario, where multitemporal images are acquired by different sensors of the same modality (see Fig. 5, HetData #1–4) and b) the multisource scenario, where multitemporal images are captured by different sensor types, such as optical and SAR (see Fig. 5, HetData #5–7 [21], [22]).

Representative events, such as flooding and wildfires, serve as critical tests for heterogeneous change detection capabilities. As shown in Table II, these datasets cover a wide range of spatial resolutions (from 0.52 to 30 m) and image dimensions (from 300 to 4135 pixels in width or height), encompassing diverse change scenarios such as flooding, construction, and lake overflow. This diversity provides a robust basis for evaluating the performance of comparison methods under varied conditions.

Fig. 5(a) and (b) present the pre-event and postevent images, respectively; (c) shows the ground truth maps highlighting structural transformations, such as newly constructed buildings; and (d) displays the corresponding scribble annotations. Please refer to Table II for further details.

2) Homogeneous Datasets: Homogeneous datasets focus on natural or human-induced changes captured under consistent sensing conditions. They are divided into two subtypes.

a) Optical datasets (see Fig. 6, HomoData #1 and 2), which include two pairs of high-resolution optical images acquired over Beijing, China, on September 30, 2012 and March 4, 2013. The spatial resolution is 1 m, and each image pair captures pre-event and postevent urban scenes to detect structural changes. The RGB images are of size 500 × 500 pixels.

b) SAR datasets (see Fig. 6, HomoData #3–6), which consist of four SAR image pairs acquired over the Yellow River Estuary, China, using the Radarsat-2 sensor. With a spatial resolution of approximately 8 m, these datasets reflect changes including farmland irrigation, river expansion, and construction.

Fig. 6(a) and (b) show the pre-event and postevent images (optical or SAR), respectively. (c) illustrates the corresponding ground truth masks, while (d) presents scribble annotations used to guide the model in detecting fine-grained, modality-specific changes. For dataset specifications, please refer to Table II.

B. Experimental Setting

To comprehensively assess the effectiveness of the proposed SHIFT framework (Base and Post versions), we conduct extensive comparisons against a wide range of SOTA methods. These competing methods are grouped into two categories based on their supervision levels and application domains: unsupervised and weakly supervised methods.

Unsupervised methods include both traditional machine learning-based approaches and recent deep learning-based

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TABLE III
QUANTITATIVE COMPARISON OF THE BINARY CHANGE RESULTS OBTAINED ON HETEROGENEOUS DATASETS (HETDATA #1–7)

Methods	HetData #1			HetData #2			HetData #3			HetData #4			HetData #5			HetData #6			HetData #7		
	OA	κ	F1	OA	κ	F1	OA	κ	F1	OA	κ	F1	OA	κ	F1	OA	κ	F1	OA	κ	F1
TransWCD	0.938	0.000	0.000	0.884	0.279	0.333	0.848	-0.002	0.000	0.956	0.000	0.000	0.967	0.444	0.457	0.964	-0.004	0.000	0.937	0.083	0.093
ACWCD	0.938	0.000	0.000	0.881	0.000	0.000	0.849	0.000	0.000	0.956	0.000	0.000	0.954	0.000	0.000	0.966	0.000	0.000	0.936	0.000	0.000
AMD-IR	0.799	0.255	0.328	0.898	0.607	0.664	0.724	0.259	0.411	-	-	-	0.950	0.572	0.597	0.686	0.115	0.170	0.782	0.171	0.254
CGAN	0.965	0.724	0.742	0.833	0.155	0.248	0.863	0.339	0.402	0.904	0.365	0.424	0.965	0.724	0.742	0.833	0.155	0.248	0.863	0.339	0.402
FPMS	0.938	0.593	0.625	0.950	0.816	0.837	0.827	0.260	0.368	0.947	0.329	0.356	0.938	0.569	0.597	0.976	0.544	0.553	0.907	0.770	0.786
X-Net	0.958	0.674	0.696	0.952	0.789	0.816	0.864	0.389	0.461	0.911	0.402	0.447	0.954	0.586	0.609	0.901	0.490	0.510	0.901	0.447	0.497
ACE-Net	0.967	0.723	0.740	0.947	0.768	0.798	0.868	0.463	0.538	0.915	0.415	0.459	0.964	0.651	0.670	0.962	0.526	0.546	0.878	0.341	0.401
SR-GCAE	0.778	0.341	0.268	0.905	0.706	0.654	0.838	0.491	0.395	0.9376	0.4246	0.4557	0.983	0.800	0.791	0.781	0.222	0.173	0.951	0.682	0.656
SDC-GAE	0.965	0.719	0.737	0.953	0.790	0.817	0.886	0.493	0.556	0.9525	0.5134	0.5378	-	-	-	-	-	-	0.977	0.795	0.806
IRG-McS	0.971	0.739	0.754	0.938	0.715	0.750	0.864	0.357	0.427	0.945	0.445	0.473	0.969	0.654	0.671	0.976	0.676	0.688	0.970	0.718	0.733
NPSG	0.910	0.415	0.460	0.753	0.194	0.323	0.737	0.135	0.290	0.841	0.232	0.286	0.939	0.563	0.591	0.976	0.638	0.650	0.775	-0.002	0.093
SCASC	0.953	0.607	0.632	0.947	0.762	0.793	0.896	0.487	0.538	0.913	0.333	0.373	0.973	0.645	0.659	0.972	0.632	0.646	0.936	0.485	0.519
GIR-MRF	0.960	0.674	0.695	0.932	0.720	0.759	0.903	0.515	0.562	0.957	0.451	0.473	0.982	0.810	0.819	0.984	0.750	0.758	0.950	0.667	0.693
SDIR	0.955	0.652	0.676	0.938	0.738	0.773	0.912	0.600	0.649	0.744	0.073	0.142	0.986	0.843	0.850	0.978	0.715	0.726	0.980	0.831	0.842
SRF	0.971	0.755	0.771	0.968	0.851	0.869	0.914	0.652	0.703	0.892	0.301	0.346	0.987	0.838	0.845	0.982	0.752	0.761	0.978	0.788	0.800
SHIFT	**0.975**	**0.783**	**0.797**	**0.977**	**0.895**	**0.908**	**0.926**	**0.701**	**0.745**	**0.957**	**0.517**	**0.539**	**0.975**	**0.729**	**0.742**	**0.983**	**0.767**	**0.776**	**0.980**	**0.824**	**0.834**
Base	+0.41%	+3.71%	+3.37%	+0.93%	+5.17%	+4.49%	+1.31%	+7.51%	+5.69%	+0.00%	+14.63%	+13.95%	-1.21%	-13.48%	-12.19%	+0.10%	+1.99%	+1.97%	+0.20%	+1.56%	+1.43%
SHIFT	**0.977**	**0.796**	**0.808**	**0.981**	**0.911**	**0.922**	**0.938**	**0.742**	**0.777**	**0.959**	**0.515**	**0.537**	**0.984**	**0.799**	**0.807**	**0.988**	**0.827**	**0.833**	**0.983**	**0.850**	**0.859**
Post	+0.62%	+5.43%	+4.80%	+1.34%	+7.05%	+6.10%	+2.62%	+13.80%	+10.53%	+0.21%	+14.19%	+13.53%	-0.30%	-4.77%	-4.50%	+0.61%	+9.97%	+9.46%	+0.51%	+2.24%	+2.02%

models. Traditional unsupervised methods, such as FPMS [43], IRG-McS [44], NPRSG [19], SCASC [45], GIR-MRF [20], SDIR [21], and SRF [22], primarily rely on handcrafted features, similarity measures, and shallow regression techniques to capture heterogeneity across modalities. In contrast, deep learning-based unsupervised methods, including AMD-IR [17], CGAN [18], X-Net [24], ACE-Net [24], SR-GCAE [23], and SDC-GAE [2], leverage encoder–decoder architectures, generative translation networks, or graph-based learning strategies to perform FA and CD.

Weakly supervised methods have recently gained attention, particularly for high-resolution building CD. To further diversify our comparisons and reflect this emerging trend, we additionally include two SOTA weakly supervised methods: TransWCD [34] and ACWCD [32]. Although both are originally designed for homogeneous, high-resolution optical datasets (e.g., LEVIR-CD), they remain representative of the current CAM-based WSCD pipeline. To our knowledge, there are currently no publicly available weakly supervised methods specifically tailored for multimodal change detection (e.g., SAR–optical). Therefore, we directly adapt the best-performing pretrained models of TransWCD and ACWCD to our multimodal datasets for further analysis.

To evaluate the effectiveness of different methods, multiple performance metrics are used, including true negatives (TNs), true positives (TPs), false negatives (FNs), false positives (FPs), overall accuracy (OA), $F1$-score ($F1$), and Kappa coefficient (κ). In the qualitative results, TN and TP are highlighted with distinct colors, while the quantitative results present the values of OA, F1, and κ.

All experiments with the proposed SHIFT framework were conducted over 400 epochs. We employed the Adam optimizer, with the learning rate initially set to $1e^{-3}$ and a higher rate of $1e^{-2}$ assigned specifically to the learnable σ parameter in the Gaussian blur module to enable rapid adaptation. Scribble annotations were automatically refined every 100 epochs, with each update expanding the coverage by up to 5%, targeting a final coverage ratio of 30%. The top 1% of high-distance pixels were added to the scribble mask after a five-iteration dilation operation, progressively enhancing the model's attention to high-confidence change areas.

C. Quantitative Comparison

Table III presents the quantitative results obtained by the proposed SHIFT and the 15 comparison methods across seven MRS datasets. Overall, SHIFT demonstrates robust performance across the majority of datasets, achieving top or near-top scores in OA, κ, and $F1$ metrics. Notably, on HetData #2, SHIFT (Post) achieves an $F1$-score of 0.922, significantly outperforming the next best method (SRF) ($F1$-score: 0.869). For HetData #5, although SHIFT does not secure the top score, it remains highly competitive with the third-highest $F1$-score. This slight difference may be attributed to the complex spatial details, particularly in building updates (from building to building), where minor overlaps between adjacent structures can lead to classification ambiguities. Meanwhile, recent weakly supervised methods, such as TransWCD and ACWCD, exhibit considerable performance drops on heterogeneous datasets, with notably lower κ and $F1$ scores, reflecting the limitations of image-level supervision when generalized to multimodal scenarios.

Table IV presents the quantitative CD results obtained by different methods on the six homogeneous datasets (two high-resolution optical datasets and four SAR datasets). In particular, on HomoData #1, SHIFT (Post) achieves the best performance across all metrics, with OA, κ, and $F1$ reaching 0.973, 0.830, and 0.844, respectively. These results show improvements of 1.78%, 26.52%, and 24.31% over the second-best method, GIR-MRF, demonstrating the effectiveness of the proposed method on high-resolution images. In addition, SHIFT (Post) achieves the highest scores for OA, κ, and $F1$ on HomoData #2–6, demonstrating its capability to refine initial predictions and handle the inherent complexities of

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TABLE IV
QUANTITATIVE COMPARISON OF BINARY CHANGE RESULTS ON HOMOGENEOUS-OPTICAL (HOMODATA #1 AND 2) AND HOMOGENEOUS-SAR (HOMODATA #3–6) DATASETS

Methods	HomoData #1			HomoData #2			HomoData #3			HomoData #4			HomoData #5			HomoData #6		
	OA	κ	F1	OA	κ	F1	OA	κ	F1	OA	κ	F1	OA	κ	F1	OA	κ	F1
TransWCD	0.960	0.680	0.701	0.988	0.663	0.668	0.819	0.000	0.000	0.989	0.000	0.000	0.967	0.000	0.000	0.941	0.000	0.000
ACWCD	0.936	0.306	0.325	0.987	0.000	0.000	0.820	0.000	0.000	0.989	0.000	0.000	0.967	0.000	0.000	0.941	0.000	0.000
IRG-McS	0.918	0.291	0.331	0.956	0.344	0.358	0.829	0.145	0.190	0.982	0.492	0.500	0.980	0.721	0.731	0.975	0.761	0.774
NPSG	0.838	0.354	0.428	0.906	0.202	0.221	0.328	0.070	0.348	0.807	0.059	0.078	0.964	0.421	0.440	0.883	0.411	0.461
SCASC	0.913	0.217	0.259	0.964	0.388	0.401	0.895	0.676	0.741	0.979	0.409	0.418	0.973	0.531	0.545	0.966	0.742	0.760
GIR-MRF	0.956	0.656	0.679	0.978	0.471	0.481	0.916	0.739	0.790	0.993	0.671	0.675	0.976	0.531	0.542	0.968	0.753	0.770
SDIR	0.907	0.461	0.511	0.951	0.320	0.335	0.941	0.798	0.834	0.984	0.527	0.535	0.984	0.740	0.748	0.975	0.803	0.817
SRF	0.936	0.526	0.560	0.965	0.409	0.421	0.935	0.766	0.804	0.990	0.651	0.656	0.985	0.771	0.778	0.973	0.795	0.809
SHIFT	**0.963**	**0.779**	**0.799**	**0.974**	**0.439**	**0.450**	**0.935**	**0.792**	**0.832**	**0.980**	**0.494**	**0.502**	**0.981**	**0.734**	**0.744**	**0.985**	**0.872**	**0.879**
Base	+0.73%	+18.75%	+17.69%	-0.41%	-6.79%	-6.44%	-0.64%	-0.75%	-0.24%	-1.31%	-26.38%	-25.63%	-0.41%	-4.80%	-4.36%	+1.03%	+8.59%	+7.59%
SHIFT	**0.973**	**0.830**	**0.844**	**0.991**	**0.695**	**0.700**	**0.949**	**0.832**	**0.863**	**0.997**	**0.872**	**0.874**	**0.987**	**0.801**	**0.807**	**0.991**	**0.918**	**0.923**
Post	+1.78%	+26.52%	+24.31%	+1.33%	+47.55%	+45.53%	+0.85%	+4.26%	+3.48%	+0.40%	+29.96%	+29.48%	+0.20%	+3.89%	+3.73%	+1.64%	+14.32%	+12.97%

SAR images. For instance, on HomoData #4, SHIFT (Post) achieves OA, κ, and $F1$ of 0.997, 0.872, and 0.874, increasing significantly the performance of the second-best method by 0.40%, 29.96%, and 29.48%, respectively. From this table, we can see that the proposed SHIFT method, particularly the SHIFT (Post) version, consistently outperforms other methods across all datasets, demonstrating its robustness and adaptability for homogeneous CD tasks. Although TransWCD and ACWCD perform relatively better on HomoData #1 and 2 due to their original design for high-resolution building CD, their generalization ability remains limited, as evidenced by the inferior results on SAR-based datasets.

D. Qualitative Comparison

Fig. 5 presents a qualitative comparison of binary change maps obtained by various methods across the seven considered heterogeneous datasets. Each row corresponds to one dataset, while columns (e)–(n) display results from different approaches. Specifically, columns (e) and (f) show the results of TransWCD [34] and ACWCD [32], which represent recent weakly supervised methods based on image-level labels. Columns (g)–(l) show the results of SOTA unsupervised MCD methods, i.e., IRG-McS [44], NPRSG [19], SCASC [45], GIR-MRF [20], SDIR [21], and SRF [22].

From the heterogeneous dataset results, one can observe that TransWCD and ACWCD [columns (e) and (f)] can roughly localize major changes in simple scenarios (HetData #2 and HetData #5). However, both methods suffer from coarse spatial resolution and significant localization errors when handling complex multimodal scenes. In contrast, methods such as IRG-McS and NPSG [(columns (g) and (h)] introduce considerable noise, leading to excessive FPs (red) that blur the structural clarity of change areas. SCASC and GIR-MRF [columns (i) and (j)] exhibit better noise suppression capability but struggle with preserving fine edge details, often resulting in fragmented or incomplete boundary detection. Our proposed SHIFT [columns (m) and (n)] significantly outperforms previous approaches. Through iterative scribble-guided feature refinement and dynamic attention mechanisms, SHIFT achieves notable reductions in both FPs and FNs (green) while maintaining clear and accurate boundaries. In particular, in complex flooding scenarios (HetData #2) and multibuilding updates (HetData #3), SHIFT demonstrates superior fine-grained localization and robustness against modality differences.

Fig. 6 shows a qualitative comparison of the results obtained by different methods on the six homogeneous datasets. Again, TransWCD and ACWCD [columns (e) and (f)] perform reasonably well on simple building CD tasks (HomoData #1 and 2), due to their design for high-resolution optical images. However, their performance deteriorates sharply in SAR-based datasets (HomoData #3–6), with substantial missing detections and inaccurate change boundaries. Meanwhile, traditional unsupervised methods, such as IRG-McS, NPSG, and GIR-MRF, also struggle to balance noise suppression and boundary preservation. The proposed SHIFT, especially after postprocessing [SHIFT (Post)], consistently identifies comprehensive and accurate change regions with minimal false detections across all datasets.

Overall, these visual comparisons confirm that weakly supervised methods relying on coarse image-level labels, like TransWCD and ACWCD, face inherent challenges in complex multimodal scenarios. In contrast, the proposed SHIFT leverages a few initial sparse yet spatially informative scribbles, achieving finer structural consistency and better spatial precision across diverse CD tasks.

E. Ablation Studies

1) Performance Analysis on Heterogeneous and Homogeneous Datasets: To evaluate the robustness and sensitivity of the proposed SHIFT model under different supervision scenarios, we conduct ablation studies across 30 scribble configurations on both heterogeneous and homogeneous datasets.

On the heterogeneous dataset (HetData, see Table V), the model demonstrates robust performance with average $F1$ scores ranging from 0.521 to 0.906. Notably, HetData #2 and 7 yield particularly stable results, with narrow 95% confidence intervals (e.g., 95% with CI given by [0.904, 0.908] and [0.803, 0.819], respectively). Overall accuracy remains consistently high across all configurations, varying between 0.921 and 0.982.

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TABLE V

COMPARISONS OF QUANTITATIVE CD PERFORMANCE UNDER DIFFERENT SCRIBBLE SETTINGS (HETEROGENEOUS DATA)

HetData	OA			κ			F1				
	Avg	Max	Min	Avg	Max	Min	Avg	Max	Min	Std	95% CI
#1	0.974	0.976	0.972	0.781	0.798	0.764	0.795	0.810	0.779	0.009	[0.791, 0.799]
#2	0.977	0.979	0.975	0.893	0.902	0.884	0.906	0.914	0.898	0.005	[0.904, 0.908]
#3	0.921	0.928	0.916	0.711	0.734	0.683	0.758	0.777	0.733	0.011	[0.753, 0.763]
#4	0.955	0.959	0.946	0.497	0.517	0.489	0.521	0.539	0.516	0.007	[0.518, 0.524]
#5	0.975	0.987	0.970	0.731	0.762	0.724	0.744	0.775	0.737	0.013	[0.736, 0.753]
#6	0.982	0.984	0.977	0.754	0.783	0.730	0.764	0.792	0.741	0.013	[0.757, 0.770]
#7	0.978	0.982	0.974	0.799	0.840	0.772	0.811	0.849	0.786	0.017	[0.803, 0.819]

TABLE VI

COMPARISONS OF QUANTITATIVE CD PERFORMANCE UNDER DIFFERENT SCRIBBLE SETTINGS (HOMOGENEOUS DATA)

HomoData	OA			κ			F1				
	Avg	Max	Min	Avg	Max	Min	Avg	Max	Min	Std	95% CI
#1	0.964	0.970	0.960	0.753	0.806	0.691	0.772	0.822	0.711	0.037	[0.755, 0.789]
#2	0.978	0.983	0.970	0.462	0.513	0.433	0.473	0.521	0.444	0.022	[0.463, 0.484]
#3	0.921	0.924	0.917	0.773	0.783	0.760	0.823	0.832	0.813	0.005	[0.821, 0.826]
#4	0.979	0.989	0.969	0.486	0.630	0.390	0.494	0.635	0.400	0.073	[0.460, 0.528]
#5	0.981	0.982	0.980	0.734	0.745	0.722	0.743	0.755	0.732	0.007	[0.740, 0.746]
#6	0.985	0.986	0.983	0.864	0.876	0.850	0.872	0.883	0.859	0.007	[0.869, 0.875]

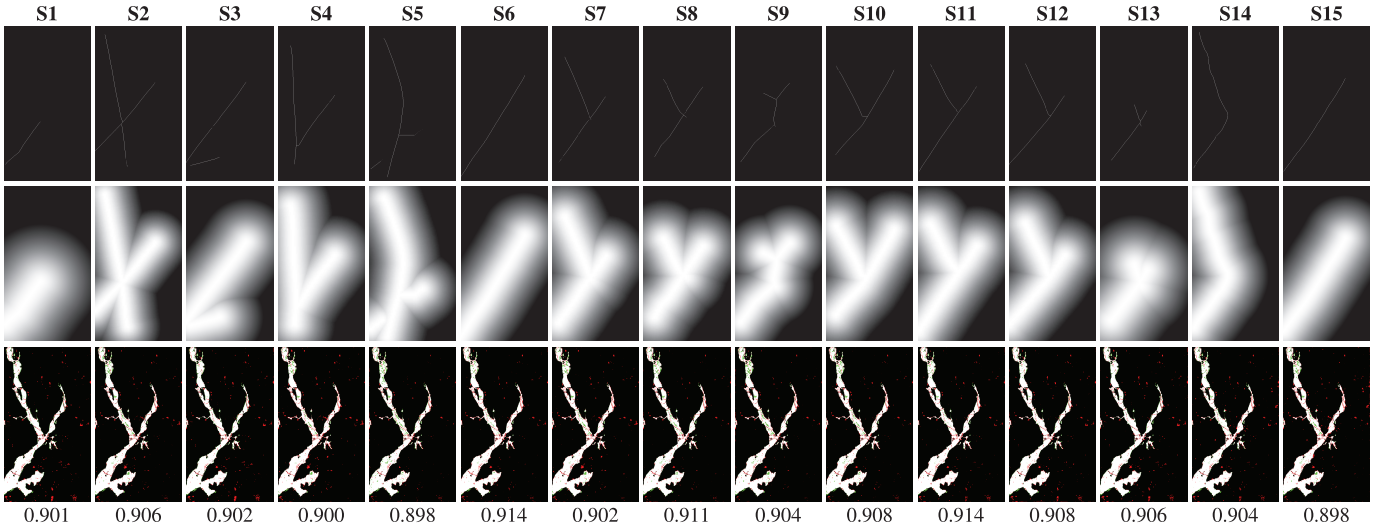


Fig. 7. Comparison of ablation study results with 15 different scribble settings on HetData #2. The first row shows different scribbles, the second row presents weight maps generated by scribbles, the third row presents binary change results generated by SHIFT (Base), and the numbers below show the values of $F1$ scores for each case.

In contrast, the ablation study on homogeneous data (HomoData, see Table VI) reveals greater performance variability. For example, HomoData #4 exhibits an F1 score range of 0.400–0.635 with a wider confidence interval ([0.460, 0.528]). However, HomoData #5 and 6 maintain more stable performance, with F1 scores consistently exceeding 0.732 and 0.859, respectively. Despite these differences, the OA values for homogeneous data remain comparable to those for heterogeneous data, ranging from 0.921 to 0.985.

Different scribble annotations inevitably affect the spatial distribution, density, and representativeness of the sparse supervision signals. Specifically, denser scribbles placed near change boundaries and covering a broader area of changed regions better guide the model's feature learning, leading to higher F1 scores, whereas sparser or less representative scribbles yield marginally lower F1 scores. In multimodal change detection, inherent cross-modal differences further amplify these effects. Nevertheless, the SHIFT model's Gaussian-guided attention and iterative refinement mechanisms effectively mitigate these issues, ensuring consistent and robust performance across varying scribble configurations. These results highlight the model's robustness to different types of input distributions and annotation sparsity, while also demonstrating its generalizability across both heterogeneous and homogeneous settings.

2) Specific Case Study: Figs. 7 and 8 illustrate a detailed ablation study on HetData #2, evaluating SHIFT's performance across 15 distinct scribble configurations (S1–S15). The visualization in Fig. 7 is organized into three components: 1) the initial scribble patterns in the first row; 2) the corresponding weight maps in the second row; and 3) the resultant binary change maps in the third row. Quantitative metrics

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TABLE VII
RUNTIME COMPARISON (SECONDS) ON MULTIMODAL DATASETS

Method	HetData							HomoData					
	#1	#2	#3	#4	#5	#6	#7	#1	#2	#3	#4	#5	#6
IRG-McS	7.42	8.05	8.02	7.93	7.55	7.38	8.11	12.24	12.18	11.53	11.46	11.54	11.30
NPSG	25.48	103.10	47.44	134.89	166.41	54.83	233.36	65.12	65.99	15.62	27.18	27.72	18.72
SCASC	7.06	3.99	3.39	3.76	3.91	3.67	3.37	9.74	5.16	5.55	5.15	3.76	3.96
GIR-MRF	25.44	68.26	46.81	47.72	20.81	34.60	72.74	24.77	29.72	16.95	12.53	12.76	11.27
SDIR	22.37	57.44	41.22	80.97	74.46	20.60	114.49	56.47	53.82	13.57	28.31	26.52	24.77
SRF	1.74	29.54	6.19	24.34	28.29	1.24	35.63	11.09	10.35	2.23	3.57	3.38	2.37
SHIFT (Ours)	12.56	54.97	25.12	44.47	54.69	10.36	60.22	25.09	25.07	8.01	12.61	13.06	9.22

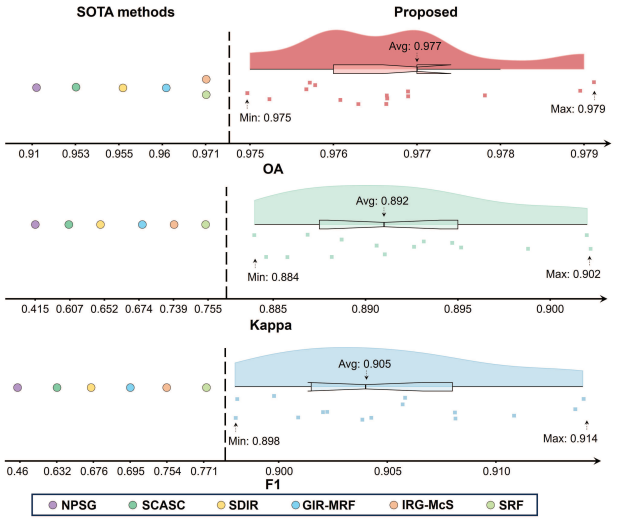


Fig. 8. Quantitative comparison of the results of the proposed SHIFT method under 15 scribble settings and of the comparative algorithms. The box plot illustrates the overall performance across all 15 scribble settings, while the scatter plots delineate the performance of the individual comparative algorithms.

and performance comparison for these scribbles are further presented in Fig. 8. From these figures, we can see that SHIFT exhibits robust performance across varying scribble inputs, as evidenced by consistently high OA, k, and F1 metrics. This consistency confirms the model's adaptability to sparse and diverse annotation patterns, ensuring reliable change detection performance even with minimal and nonspecific annotations.

F. Computational Efficiency Analysis

To evaluate the computational efficiency of SHIFT, we report the total computational time (in seconds) across both heterogeneous and homogeneous datasets to assess its efficiency under different data modalities. The detailed results are summarized in Table VII.

SHIFT was implemented in Python using PyTorch and evaluated on a server equipped with an NVIDIA RTX 3060 GPU (12-GB VRAM), an Intel Core i9-12900H CPU, and 32-GB RAM. In contrast, baseline methods (IRG-McS, NPSG, SCASC, GIR-MRF, SDIR, and SRF) were implemented in MATLAB R2018a and executed on a CPU-only local machine (also Intel Core i9-12900H with 32-GB RAM). Although differences in hardware and programming environments limit direct comparison, the results yield insights regarding the relative computational demands of the methods.

As shown in Table VII, SHIFT demonstrates a balanced tradeoff between accuracy and computational cost, with computational times ranging between 10.36 and 60.22 s for HetData #1–7 and 8.01–25.09 s for HomoData #1–6. While SCASC and SRF methods exhibit shorter computational times for specific scenarios, SHIFT consistently outperforms computationally intensive methods such as NPSG and SDIR, particularly in challenging scenarios (e.g., HetData #2, 4, and 7).

Overall, SHIFT maintains stable computational performance across diverse datasets, highlighting its practical viability for MRS-CD. Importantly, all evaluated methods operate within acceptable timeframes, typically requiring less than 2 min/image pair, underscoring their suitability for real-world applications.

V. CONCLUSION

This article introduced SHIFT, a weakly supervised framework for MRS-CD utilizing sparse scribble annotations as supervision. By integrating an iterative refinement mechanism with a Gaussian blur module, SHIFT effectively directs attention toward potential change regions, reducing reliance on dense annotations while maintaining high detection accuracy. Extensive experiments across 13 public datasets, encompassing diverse scenarios and sensor types, validated the robustness of the proposed method. SHIFT consistently outperforms traditional machine learning and deep learning-based approaches, achieving superior performance in both quantitative metrics and qualitative assessments. Ablation studies further demonstrate SHIFT's adaptability to varying scribble annotations, underscoring its suitability for real-world scenarios where detailed annotations are costly or impractical. Moreover, computational efficiency analyses confirm that SHIFT achieves an optimal balance between inference speed and detection accuracy, highlighting its potential for deployment in resource-constrained environments.

However, certain limitations of the proposed method should be acknowledged. While SHIFT incorporates an iterative mechanism to dynamically refine attention based on initial scribble inputs, its performance remains sensitive to the placement and density of these initial seed annotations. In particular, scenarios involving extremely subtle or small-scale changes may benefit from more carefully distributed scribble cues to ensure adequate spatial coverage. Furthermore, while iterative refinement enhances detection precision, it incurs moderate computational overhead compared with single-pass models.

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Future work will investigate robust annotation-guided initialization strategies, explore model compression for efficiency, and extend SHIFT to complex multimodal fusion tasks and real-world deployment.

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Yongjie Zheng (Student Member, IEEE) received the B.S. degree in remote sensing science and technology from Henan Polytechnic University, Jiaozuo, China, in 2018, and the M.S. degree in photogrammetry and remote sensing from Tongji University, Shanghai, China, in 2021. She is currently pursuing the Ph.D. degree in remote sensing image analysis with the University of Trento, Trento, Italy.

Her research interests include deep learning, feature extraction, feature fusion, and remote sensing image classification and change detection.



Sicong Liu (Senior Member, IEEE) received the B.Sc. degree in geographical information system and the M.E. degree in photogrammetry and remote sensing from China University of Mining and Technology, Xuzhou, China, in 2009 and 2011, respectively, and the Ph.D. degree in information and communication technology from the University of Trento, Trento, Italy, in 2015.

He is currently an Associate Professor with the College of Surveying and Geo-Informatics, Tongji University, Shanghai, China. His research interests include multitemporal data analysis, change detection, and multispectral/hyperspectral remote sensing in Earth observation and planetary exploration.

Dr. Liu has been serving as a Program Committee Member for SPIE Remote Sensing Symposium: Image and Signal Processing for Remote Sensing since 2020. He was a Winner (ranked as third place) of Paper Contest of the 2014 IEEE GRSS Data Fusion Contest. He is the Technical Co-Chair of the Tenth International Workshop on the Analysis of Multitemporal Remote Sensing Images (MultiTemp 2019). Since 2017, he has been serving as the Session Chair for many international conferences, such as International Geoscience and Remote Sensing Symposium. He is an associate editor and a guest editor of several international journals.



Lorenzo Bruzzone (Fellow, IEEE) received the Laurea (M.S.) degree (summa cum laude) in electronic engineering and the Ph.D. degree in telecommunications from the University of Genoa, Genoa, Italy, in 1993 and 1998, respectively.

He is currently a Full Professor of telecommunications with the University of Trento, Trento, Italy, where he teaches remote sensing, radar and digital communications. He is the Founder and the Director of the Remote Sensing Laboratory, Department of Information Engineering and Computer Science, University of Trento. He is the Principal Investigator of many research projects. Among the others, he is currently the Principal Investigator of the Radar for Icy Moon Exploration (RIME) and of the Subsurface Radar Sounder (SRS) instruments in the framework of the JUICE and the Envision missions, respectively, of European Space Agency (ESA) and is the Science Lead of the High Resolution Land Cover project in the framework of the Climate Change Initiative of ESA. He is the author (or a co-author) of more than 420 scientific publications in refereed international journals, more than 390 papers in conference proceedings, and 25 book chapters. He is the editor/co-editor of 18 books/conference proceedings and two scientific books. His papers are highly cited, as proven from the total number of citations (more than 59 000) and the value of the H-index (114) (source: Google Scholar). He was invited as a keynote speaker at more than 40 international conferences and workshops. His research interests are in the areas of remote sensing, radar and SAR, signal processing, machine learning, and pattern recognition. He promotes and supervises research on these topics within the frameworks of many national and international projects related to Earth observation and planetary exploration.

Dr. Bruzzone was a member of the Administrative Committee of the IEEE Geoscience and Remote Sensing Society (GRSS) from 2009 to 2023, where he has been the Vice-President for Professional Activities since 2019. He was ranked in the First Place in the Student Prize Paper Competition of the 1998 IEEE International Geoscience and Remote Sensing Symposium (IGARSS), Seattle, in July 1998. Since that, he was a recipient of many international and national honors and awards, including the recent IEEE GRSS 2015 Outstanding Service Award, 2017 and 2018 IEEE IGARSS Symposium Prize Paper Awards, the 2019 WHISPER Outstanding Paper Award, and the 2022 Letter Prize Paper Award for the best paper published on the IEEE GEOSCIENCE AND REMOTE SENSING LETTERS. He was a Distinguished Speaker of the IEEE Geoscience and Remote Sensing Society from 2012 to 2016. He is the Co-Founder of the IEEE International Workshop on the Analysis of Multi-Temporal Remote-Sensing Images (MultiTemp) series and is currently a member of the Permanent Steering Committee of this series of workshops. Since 2003, he has been the Chair of the SPIE Conference on Image and Signal Processing for Remote Sensing. He has been the Founder of *IEEE Geoscience and Remote Sensing Magazine* for which he was the Editor-in-Chief from 2013 to 2017. He was a guest co-editor of many special issues of international journals. He is an Associate Editor of IEEE TRANSACTIONS ON GEOSCIENCE AND REMOTE SENSING.

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